User Retention Analysis

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Context

Context: The PM has identified what they think are the key features in our product. The PM intuition is that if a user uses one of those features in the first week of joining, they will be retained longer.

Aim: The aim of this piece of work is to prove/disprove the product manager's hypothesis and come up with other findings based on the data.

Data: The dataset used for this analysis contains weekly activity from users across their organisation type, actions performed, device used etc. between May to October 2025. Some data quality issues were detected in the data such as:

- NULL values in ORGANISATION_CARE_CLASSIFICATION
- Outliers detected in TOT_ACTIVITY_WEEK

Context

1 **Key Metrics and Dimensions:** Defining key activation and retention metrics

2 **About the Users:** A snapshot of the user activity

3 **Retention:** Defining time buckets for retention and the rationale behind them

4 **Impact of Activation on Retention**

5 **Impact of Other Signup Behaviour**

6 **Findings**

7 **Appendix**

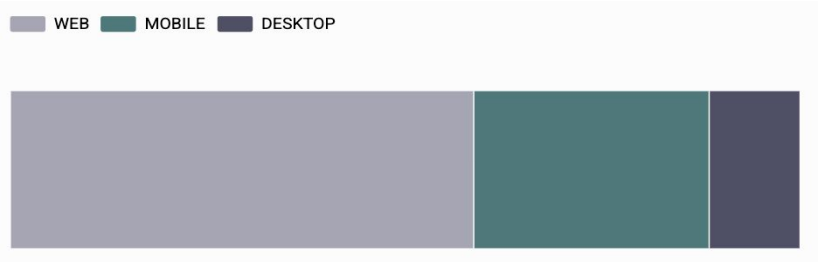
Key Metrics and Dimensions

Metric/Dimension	Business Definition
Activation Status	‘Activated’ refers to users that have performed a key activity as defined by the PM in week 1 - this week is referred to as ‘signup week’ throughout the analysis.
Key Activity Level	Refers to the level of key activity per week because of number of key actions done by the user
Key Activity Type	Refers to the type of key activity such as messaging a clinician, patient or sending a video link to a patient
Retention Rate	The percentage of users who remain active after a specified period among those who had sufficient time since signup to reach that period. Active refers to any actions (other or key) made by the user.

About the Users

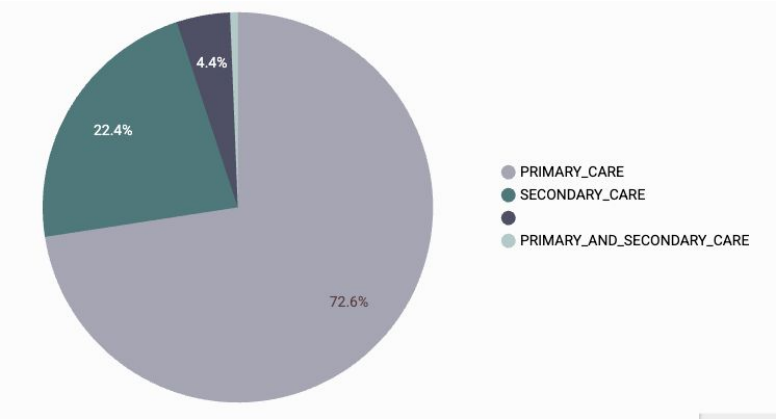
Total 25,897 Users across 6,608 Organisations

Majority of activity happens on the web platform



'Activated' user cohort forms only a quarter of all users

% Users activated in week 1
25.71%



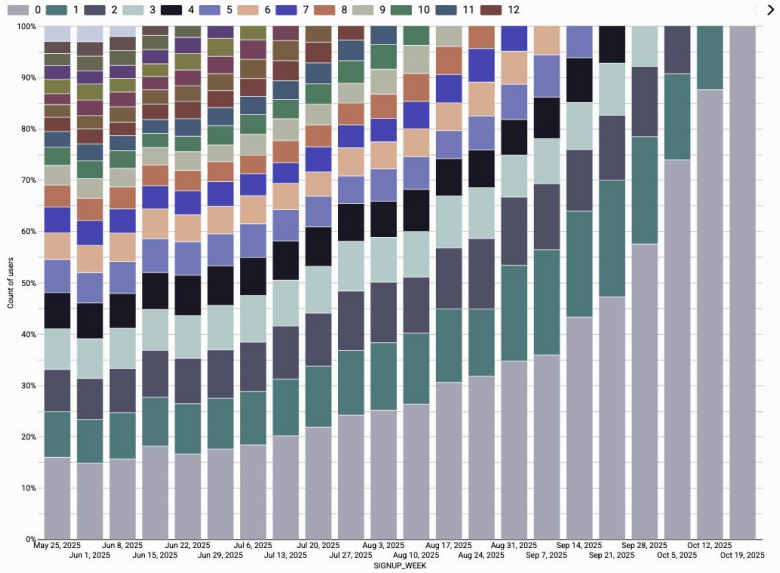
Majority of users are accessing through Primary Care Organisations

Retention

Although the dataset spans 20+ weeks, retention analysis is limited to 12 weeks due to declining eligibility for later periods. Users join continuously throughout the May–October window, meaning later cohorts have not had sufficient time to reach longer retention milestones (see chart on the right). Beyond 12 weeks, the eligible sample size becomes too small to support reliable conclusions.

Retention Rate has been calculated in the following buckets:

2 weeks	to indicate immediate return activity and understand user behaviour
4 weeks	Short - term retention
8 weeks	Medium-term retention
12 weeks	Medium to long term retention
16 weeks	Long term retention



Count of users by signup week, broken down by weeks since signup

Impact of Activation on Retention

While users not activated in week 1 show strong initial return activity, activated users have a more stable and sustained retention over the long term. However, the uplift in retention does not seem significantly higher.

This initial insight somewhat supports the PM hypothesis but we're going to investigate impact of some factors.

		WEEK 2		WEEK 4		WEEK 8		WEEK 12		WEEK 16	
ACTIVATION STATUS	ALL USERS	ELIGIBLE USERS	RETENTION RATE	ELIGIBLE USERS	RETENTION RATE	ELIGIBLE USERS	RETENTION RATE	ELIGIBLE USERS	RETENTION RATE	ELIGIBLE USERS	RETENTION RATE
NOT ACTIVATED	20600	20561	50.14%	20255	35.37%	16650	19.62%	12496	14.97%	7853	12.56%
ACTIVATED	5297	5294	26.92%	5271	25.35%	4573	22.72%	3455	20.78%	2113	18.69%

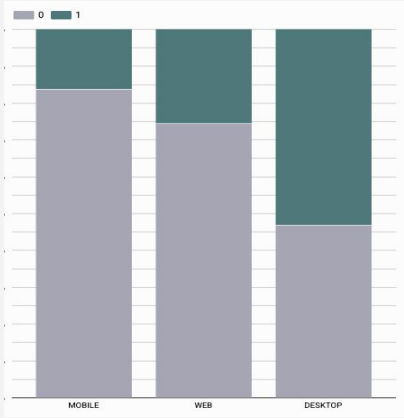
Week 4 Retained | Week 4 Not Retained

Impact of Other Signup Behavior

The above charts show week 4 retention only but similar trends can be observed longer term

Signup Platform

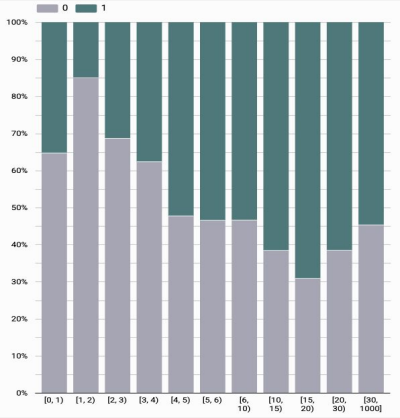
Users that sign up on desktop have longer retention even though they make up a small proportion of users



User count by platform by week 4 retention

Signup Key Activity Intensity

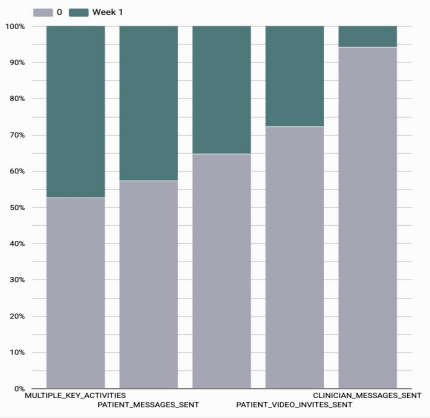
Retention trends are a lot clearer when observing number of key actions done in week 1, not just whether a key action was done. Retention increases for users that perform more than 1 key action in sign up week.



User count by signup key activity count by week 4 retention

Signup Key Activity Type

Sending patient messages or doing multiple activities in signup week is linked to stronger retention



User count by signup key activity by week 4 retention

Findings

Hypothesis: While activated users have more stable retention longer term, the overall increase is not significant enough to prove the PM hypothesis. The PM's definition of activation also accounts for a small cohort (25%) of users only.

Initial analysis indicates retention is driven more by intensity and context of activation, than by just completing a key action

In order to prove this hypothesis, I would do the following:

- **Define Activation Metric to include intensity** as retention becomes stronger for users that do more than one key action in sign-up week. Binary activation is not a strong factor on its own.
- **Explore 'Other' actions for potentially valuable 'Key' activities:** Users that performed key activities form a small proportion of all users. We might be losing valuable insights bucketed as 'Others'
- **Understand impact of other factors:**
  Platform: Users have much stronger and longer retention if they used desktop in their first week even though they form a smaller proportion of users
  Organisation type: Users that perform a key action with a primary care organisation in their first week have really strong retention, although this is again a really small cohort hence not impacting overall retention significantly
- In terms of next steps, it would also be beneficial to build at retention as a dynamic metric instead of hard coded values

Thank you

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